



Unsolved mysteries

Dimensionless constants, free will, abundance of matter, and time... know about what we don't know about. <https://digit.in/apr20-63>



Unboiling an egg

Scientists have managed to partially unboil an egg, without breaking any laws of physics. <https://digit.in/apr20-64>

THE ARROW OF TIME

Physicists are at a loss to explain why time seems to have a direction.

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In colloquial usage, the past is considered to be behind us and the future ahead of us. Personally, we can

remember the past and only guess at what the future holds. In that sense, time can be said to have a direction, going from the past to the future. When it comes to the entirety of civilisation, we have records of the past, but none of the future. Predicting the future is possible to some extent, but not accurately and entirely. Most physical processes also have a directionality. While an egg can be boiled, it is difficult to unboil an egg. This directionality of time in scientific terms can be called “asymmetry”, that is time is not the same in both the directions.

The recognition of the directionality of time by our individual and collective consciousness, is something that is very important when it comes to understanding time. In 1927, the English astrophysicist Arthur Stanley Eddington came up with the concept of the “arrow of time”, in a popular science book called *The Nature of the Physical World*. The book was written just when our understanding of the universe was changing dramatically, with the splitting of the atom, the advent of quantum theories, and Einstein publishing his theory of relativity. The concept of time’s arrow was actually introduced for the general public, but has been adopted by physicists. Additionally, the concept was introduced as a philosophical property, not a physical one, noting that the existence of the arrow of time “is equally insisted on by our reason-

ing faculty, which tells us that a reversal of the arrow would render the external world nonsensical.” While it is intuitive that time seems to have a direction, Einstein’s theory of relativity is not at all intuitive. This is because, according to Einstein, time and space are smushed up together to form a single entity, the spacetime continuum. However, while we can physically see space, and even move about it, we cannot see time, and we can only move forward in it. In fact, the four dimensional universe proposed by Einstein faced early opposition precisely because it seemed to overlook the directionality of time.

Another common sense perception is that time seems to move in lockstep for everyone. Supposing an astronaut says goodbye to his grandfather, and takes a trip to Mars and comes back, and then greets his grandfather